

ANALYSTLAB AFRICA

EXPERIENCE LAB INTERNSHIP PROGRAMME

HEALTHCONNECT CLINIC EXPERIENCE LAB

WEEK 6 — PROJECT STATUS UPDATE

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Field	Detail
Project:	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Programme:	AnalystLab Africa Experience Lab
Function:	Project Management
Reporting Period:	Week 6
Reporting Stage:	Integration → Advanced Development → Validation
Status Date:	Week 6 Review
Document Version:	1.0
Overall Project Status:	AMBER — CONTROLLED INTEGRATION
Status Owner:	Project Management

1. EXECUTIVE PROJECT POSITION

Week 6 represents a transition from predominantly parallel workstream development toward controlled cross-workstream integration and validation preparation.

All four technical workstreams have produced substantive evidence during the reporting period. Data Analytics strengthened the project's evidence base through statistical validation and interaction analysis; Data Science advanced the predictive-development pathway; ML Engineering demonstrated a technically functioning scoring and integration pathway; and Generative AI progressed into scenario-based behavioural testing.

The project remains AMBER because progress is not yet uniform across the integration chain.

The principal exposure is the incomplete transfer of the intended Data Science candidate model into the ML implementation pathway. Additional conditions remain around downstream incorporation of analytical findings, operational interpretation of ML outputs, GenAI behavioural consistency and final treatment of Cancelled records within the predictive target.

There is currently no evidence of a project-wide blocker. However, the absence of a blocker should not be interpreted as closure of the underlying dependencies.

Executive Determination

The project is sufficiently progressed to enter Week 7 Testing & Refinement under controlled conditions, but the evidence does not yet support a claim of fully integrated or solution-ready status.

The immediate management requirement is therefore to close material interfaces and strengthen acceptance evidence rather than initiate additional parallel development.

2. PROJECT HEALTH DASHBOARD

Control Dimension	RAG	Current Position	Management Interpretation
Scope	Green	Core problem and solution direction remain unchanged.	No material scope deviation established.
Schedule	Grey	No Week 6 schedule re-baselining or timeline update was undertaken by PM.	Schedule variance is not reassessed in this report.
Data & Analytical Evidence	Green	Analytical evidence materially strengthened.	Suitable for controlled downstream use, subject to interpretation boundaries.
Predictive Development	Amber	Development advanced, but comparative model evidence and final candidate selection remain outstanding.	Conditional progression.
Technical Integration	Amber	ML scoring and interface capability demonstrated; intended DS candidate integration pending.	Material interface remains open.
Generative AI	Amber	Scenario testing established useful evidence but exposed behavioural inconsistency.	Refinement and retesting required.
Cross-Track Coordination	Amber	Material exchanges occurred, but integration maturity varies by interface.	Active integration control required.
Quality / Evidence Sufficiency	Amber	Evidence quality improved, but validation maturity is uneven.	Avoid premature solution-level claims.
Risk Exposure	Amber	No Critical exposure established; several High/Medium exposures remain under control.	Continued monitoring and targeted closure required.
Operational Readiness	Amber	Technical components have progressed ahead of accepted operational use conditions.	Not yet solution-ready.
Overall Project	AMBER	Controlled progress with material integration conditions.	Proceed with conditions into Week 7.

RAG legend: ☐ Green = controlled/on plan | ☐ Amber = material attention or condition | ☐ Red = material threat requiring escalation | ☐ Grey = not assessed / not applicable for this reporting period.

3. REPORTING-PERIOD PERFORMANCE

3.1 What materially changed during Week 6

Data Analytics

The analytical workstream advanced from primarily descriptive exploration to a more statistically supported evidence base.

Material developments included:

- Statistical testing of key no-show relationships;
- Interaction analysis across relevant patient and appointment dimensions;
- Validation of the Previous No-Show Rate measure;
- Refinement of the distance interpretation as an associated logistical signal rather than causal evidence;
- Removal of waiting time as an established predictive driver; and
- Further development of decision-support dashboard outputs.

Status: Green for Week 7 testing/refinement.

Data Science

Data Science progressed toward a binary Attended vs No-Show predictive formulation and developed the supporting feature and modelling pipeline.

Work completed or prepared included:

- Engineered predictive features;
- Revised Logistic Regression development;
- Random Forest development;
- XGBoost preparation;
- hyperparameter-search preparation;
- SMOTE preparation;
- Threshold-tuning preparation; and
- False-positive / false-negative analysis preparation.

The critical limitation is that the planned comparative evaluation had not yet been fully executed against the dataset.

Therefore: methodological progress is established; improved predictive performance and final candidate selection are not yet established.

Status: Amber.

Machine Learning / ML Engineering

ML advanced materially from a training-oriented pipeline toward technically demonstrated scoring capability.

Week 6 evidence includes:

- Correction of the silent target-building error;
- Correction of the model-wrapper truthiness defect;
- Schema alignment for missing one-hot categories;
- End-to-end batch scoring;
- Scoring against a 5,000-row dataset and 60-row simulated upcoming batch;
- Output and schema validation;
- Reproducibility checks; and
- 38/38 integration tests passing.

The intended Data Science candidate model was not available for the actual integration exercise. A Random Forest stand-in was used to verify the technical pathway.

Status: Amber — technical pathway demonstrated; intended cross-track integration pending.

Generative AI

GenAI progressed into live scenario testing across defined behavioural and safety categories.

Evidence was obtained for selected behaviours involving:

- Diagnosis-related refusal;
- Emergency escalation;
- Pricing redirection;
- Avoidance of invented pricing; and
- Treatment/medication boundaries.

Testing identified an inconsistent emergency-disclaimer response.

The model-alias change necessitated by retirement of the previous Gemini model is treated as a compatibility/testing change and not as evidence of model performance improvement.

Status: Amber — conditional progression into Week 7 refinement.

4. DELIVERY STATUS BY WORKSTREAM

Workstream	Week 6 Outcome	Evidence Maturity	Current Status	Principal Carry-Forward
Data Analytics	Statistical validation, interaction analysis and refined decision-support evidence completed.	Strong	Green	Demonstrate downstream incorporation where applicable.
Data Science	Predictive formulation and model-development pathway advanced.	Developing	Amber	Execute comparative evaluation and select candidate.
ML Engineering	Scoring pathway, interface controls and integration testing demonstrated.	Strong technically	Amber	Integrate actual DS candidate and define operational use.
Generative AI	Scenario testing completed with specific behavioural issue identified.	Scoped	Amber	Retest disclaimer behaviour and broaden relevant evidence.

5. SCOPE AND REQUIREMENTS POSITION

Scope Position: GREEN

No material change to the core project problem or stated solution direction was established during Week 6.

The project continues to address:

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI.

The current technical direction remains consistent with the project objective:



Scope Control Observation

The move from three-class to binary predictive modelling represents a modelling-definition decision within the current predictive workstream, rather than a confirmed project-scope expansion.

However, treatment of Cancelled appointments remains insufficiently resolved for complete target-definition closure and should be explicitly documented before predictive integration is treated as final.

6. SCHEDULE POSITION

Schedule Status: GREY — NOT REASSESSED IN WEEK 6

The Project Management function did not undertake a Week 6 update or re-baselining of the Project Timeline.

Accordingly, this status report does not make a new claim regarding:

- Schedule variance;
- Milestone slippage;
- Critical-path movement;
- Earned schedule performance; or
- Revised completion dates.

This is intentional.

The current report instead assesses integration readiness and Week 7 entry conditions based on available Week 6 evidence.

Schedule Control Implication

The absence of a new schedule assessment should not be interpreted as evidence of schedule compliance. Schedule position remains a separate control requiring formal review when the project timeline is next assessed.

7. INTEGRATION HEALTH

Integration Status: AMBER

Week 6 established substantive coordination across the four technical workstreams, but interface maturity is uneven.

Interface	Current Evidence	Status
Analytics → DS	Findings/features reportedly exchanged; downstream incorporation not fully traceable.	Conditional
DS → ML	Intended candidate not transferred; technical pathway tested with stand-in.	Open interface
ML → Operational Use	Scoring technically demonstrated; operational interpretation undefined.	Conditional
GenAI → PM Governance	Testing evidence incorporated into project controls.	Controlled
GenAI → Validation	Selected scenario evidence available; broad reliability not established.	Conditional

Integration Finding

The project has demonstrated cross-track coordination, but this should not be represented as completed solution-level integration.

The most material unresolved interface is the DS → ML candidate-model handoff.

8. QUALITY AND VALIDATION POSITION

Quality Status: AMBER

The quality position improved during Week 6 primarily through stronger evidence generation and clearer interpretation boundaries.

Established

- Statistically supported analytical findings;
- Technically tested ML scoring capability;
- Schema and output validation;
- Reproducibility evidence;
- Defined GenAI scenario evidence;
- Documented interface conditions.

Not Yet Established

- Superior predictive performance of the revised DS candidates;
- Final predictive candidate selection;
- Actual integration of the intended DS candidate into ML;
- Accepted operational meaning of ML risk scores;
- Broad GenAI behavioural reliability;
- Intervention effectiveness; and
- Complete solution-level validation.

Quality Control Position

The principal concern is evidence sufficiency, not lack of technical activity.

The project should therefore maintain the following evidence hierarchy:

Development Evidence → Integration Evidence → Validation Evidence → Acceptance Evidence → Readiness

A completed development activity should not automatically be reported as validated or operationally ready.

9. RISK AND ISSUE POSITION

Overall Exposure: AMBER

No Critical project risk is established from the Week 6 evidence reviewed.

Material active exposures remain concentrated around:

1. DS candidate-model transfer and integration;
2. Predictive model comparison and selection;
3. Operational interpretation of ML outputs;
4. GenAI disclaimer consistency;
5. Downstream traceability of analytical findings; and

6. Unresolved treatment of Cancelled records.

These are controlled through the existing project registers and are not reproduced here in full.

Status Interpretation

The project is not blocked, but several dependencies remain consequential to the quality and completeness of Week 7 integration.

10. DECISION POSITION

The following Week 6 decisions have direct implications for current project execution:

Decision	Current Position	Consequence
Predictive target	Binary Attended vs No-Show adopted for current development.	Supports current modelling direction; Cancelled treatment remains open.
Waiting time	Not retained as an established predictive driver.	Prevents unsupported use of the variable as a validated risk signal.
Distance	Treated as an associated/logistical signal rather than causal evidence.	Protects analytical interpretation.
No Reminder	Retained as a meaningful category.	Preserves informational content of the dataset.
GenAI model alias	Updated following model retirement.	Restores test compatibility; does not establish performance improvement.
ML stand-in	Used to test technical integration pathway while DS candidate was unavailable.	Provides pathway evidence without closing DS→ML integration.
Technical vs operational readiness	Explicitly separated.	Prevents scoring capability being represented as operational readiness.
Final predictive candidate	Pending.	Prevents premature model-selection closure.

11. RESOURCE AND CAPACITY POSITION

Resource Status: GREEN / CONTROLLED

The four technical workstreams remain represented within the current project structure.

No material resource shortfall has been evidenced during the reporting period.

The principal capacity concern is therefore coordination dependency rather than headcount.

The most significant example is the outstanding DS→ML handoff: the constraint is the availability and acceptance of the required technical output, not an identified shortage of personnel.

12. STAKEHOLDER / GOVERNANCE POSITION

Governance Status: GREEN

Project Management has consolidated workstream evidence through:

- Cross-track coordination control;
- Integration and dependency monitoring;
- Issue control;
- Decision control; and
- Readiness assessment.

No material governance failure is established from the Week 6 evidence.

The principal governance requirement for Week 7 is maintaining evidence traceability and disciplined status language, particularly where workstream progress could otherwise be interpreted as completed integration or validation.

13. MATERIAL VARIANCES / EXCEPTIONS

A formal project variance should only be recorded where a baseline, approved requirement, milestone or acceptance condition can be compared against actual performance.

Based on the evidence available:

Area	Variance Position	Comment
Scope	None established	Core problem and solution direction remain unchanged.
Schedule	Not assessed	No Week 6 timeline review/re-baselining undertaken.
Cost	Not assessed	No cost baseline or Week 6 financial evidence available in the PM record.
Technical Integration	Material condition	DS candidate handoff remains incomplete.
Predictive Development	Evidence gap	Comparative performance and final candidate selection remain outstanding.
Validation	Controlled gap	Evidence maturity varies by workstream.
Operational Readiness	Conditional	Technical capability has progressed ahead of accepted operational use.

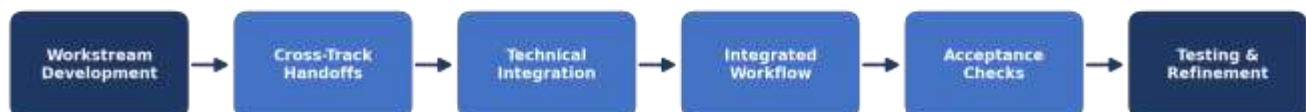
This prevents an absence of evidence from being incorrectly reported as either compliance or variance.

14. FORECAST AND OUTLOOK

Near-Term Forecast: CONTROLLED WITH CONDITIONS

The project is expected to progress into Week 7 Testing & Refinement provided the material interfaces remain actively managed.

The immediate trajectory is:



The project is currently between cross-track handoff/integration and integrated validation.

Forecast Confidence

Moderate.

The principal uncertainty is concentrated in the outstanding DS→ML handoff and the evidence required to convert component-level technical progress into accepted integrated behaviour.

No evidence currently indicates that the project trajectory requires a fundamental change in scope or solution direction.

15. WEEK 7 OBJECTIVES

Objective	Expected Outcome	Success Condition
Predictive Integration	Actual DS candidate integrated into ML pathway.	Candidate received, accepted and technically verified.
Model Selection	Evidence-based candidate selection.	Comparative results and selection rationale documented.
Analytical Traceability	Analytics contribution demonstrably incorporated downstream.	Resulting model/feature change identifiable.
Operational Scoring	ML outputs assigned accepted operational meaning.	Decision conditions documented and accepted.
GenAI Refinement	Known behavioural inconsistency retested.	Expected disclaimer behaviour demonstrated across relevant scenarios.
Target Definition	Cancelled-record treatment resolved.	Predictive target definition formally documented.
Integrated Testing	Components prepared for controlled validation.	Testable outputs, limitations and acceptance conditions available.

16. MANAGEMENT ATTENTION REQUIRED

The following matters warrant active management attention entering Week 7:

1. Close the DS→ML interface

The intended candidate model should be transferred and verified rather than continuing to rely on a stand-in for integration evidence.

2. Protect evidence integrity

Analytical significance, technical test success and model-development progress should not be converted into claims of causality, predictive superiority or solution readiness without supporting evidence.

3. Define operational use

ML scoring capability requires an accepted interpretation before it can become an operational intervention input.

4. Retest GenAI behaviour

The identified disclaimer inconsistency should be treated as a targeted refinement and regression requirement.

5. Maintain interface ownership

Open coordination actions should retain clear provider/receiver ownership so that integration gaps do not become ambiguous shared responsibilities.

17. OVERALL PROJECT ASSESSMENT

AMBER — CONTROLLED INTEGRATION

The project has made substantive progress during Week 6 and has established a stronger evidence base for the transition into testing and refinement.

The evidence supports the following conclusions:

- Data Analytics has materially strengthened the analytical foundation.
- Data Science has advanced predictive development but has not yet demonstrated final model superiority or selection.
- ML Engineering has demonstrated a technically capable scoring pathway and resolved key interface defects.
- Generative AI has progressed into scenario testing and generated actionable behavioural evidence.
- Cross-track coordination has become more structured and traceable.
- No project-wide blocker has been established.

At the same time, the project should not yet be described as fully integrated, fully validated or operationally ready.

The outstanding conditions are concentrated at identifiable interfaces rather than representing a general failure of project execution.

Final PM Determination

Proceed to Week 7 Testing & Refinement under controlled integration conditions, with priority given to completion of the DS→ML handoff, evidence-backed candidate selection, downstream traceability, operational interpretation of ML outputs and targeted GenAI refinement.

18. STATUS REPORT GOVERNANCE

Prepared by: Project Management

Reporting Basis: Verified Week 6 workstream evidence and controlled PM records

Primary Supporting Records:

- Cross-Track Coordination Record
- Integration & Dependency Register
- Integration Issue Log
- Decision Log
- Risk & Dependency Register

Status Method: Evidence-based RAG assessment

Next Review Point: Week 7 Testing & Refinement Review

Status Authority: Project Management, subject to Project Leadership review where escalation is required

Where evidence is incomplete, this report records the position as conditional, pending evidence, not assessed or not established, rather than inferring completion.

19. VERSION CONTROL

Version	Reporting Stage	Change
1.0	Week 6	Initial controlled Project Status Update consolidating workstream, integration, quality, risk, decision and readiness positions.
1.1	Week 7	Updated following Week 7 testing/refinement activity and closure or movement of Week 6 carry-forward actions.
2.0	Post-Week 7 Review	Re-baselined following formal assessment of integration, validation and readiness evidence.